

Cybersecurity in the Oil and Gas Sector: Vulnerabilities, Solutions, and Future Directions

Prasad Pothana

*Energy and Petroleum Engineering
University of North Dakota
Grand Forks, ND USA
prasad.pothana@und.edu*

Vasanth Gokapai

*Energy and Petroleum Engineering
University of North Dakota
Grand Forks, ND USA
vasanth.gokapai@und.edu*

Ananth N. Ramaseri-Chandra

*Computer Science, Academic Success
Turtle Mountain College
Belcourt, North Dakota, USA
aramaserichandra@tm.edu*

Abstract—The integral role of the oil and gas (O&G) industry in the global economy and digital ecosystem exposes it to escalating cyber threats that target critical operational technology and information systems. This literature review comprehensively examines the sector's cybersecurity complexities, drawing from academic articles, industry reports, and case studies. Critical vulnerabilities exposing the industry to cyber-attacks are identified, and existing cybersecurity measures implemented by companies to combat these threats are evaluated. The review categorizes industry challenges into technological, organizational, and regulatory segments. Integrating legacy systems with modern technologies like cloud computing and IoT presents challenges. Organizationally, the focus is on cybersecurity culture, including employee training and policy enforcement. Regulatory challenges stem from the compliance burdens of national and international cybersecurity standards. The review discusses cybersecurity solutions, emphasizing innovative technologies like artificial intelligence for efficient risk detection and response. Strategic measures, such as incident response plans and executive leadership, are highlighted in fostering cybersecurity awareness. Future challenges driven by technological advancements and evolving cyber threat tactics are explored, with recommendations for ongoing research areas to strengthen the industry's cyber defenses and resilience. This review serves as a critical resource for stakeholders, policymakers, and researchers, underscoring the urgent need for proactive cybersecurity strategies to safeguard the (O&G) industry's assets and infrastructure.

Index Terms—Cybersecurity, Critical Infrastructure, Regulatory Compliance, Technological Vulnerabilities, Oil and Gas, Energy

I. INTRODUCTION

This increased reliance on digital technologies globally makes security critical for all industries, including critical infrastructure like the (O&G) industry. The information technology and operational technology (IT and OT) components required for modernizing operations also create new entry points for cyber attackers [1], [2]. Cyber incidents can cause shutdowns of operations, endanger human life, and cause significant economic and reputational damage to the O&G industry [3]. A selected list of major cyber attacks that occurred in the energy sector recently is presented in Table I.

Despite the critical nature of these threats, the O&G sector faces unique challenges in cybersecurity management. These challenges stem from several factors: the sector's global distribution of assets, the mix of legacy and modern technologies,

and the varying levels of cybersecurity maturity across different companies [4]. Moreover, regulatory frameworks governing cybersecurity in the industry are often fragmented and vary significantly by region, complicating compliance efforts [5].

Research to date has extensively documented various cyber threats specific to the O&G industry, ranging from targeted phishing attacks to sophisticated state-sponsored intrusions [6]. However, there needs to be more literature regarding reviews that connect these threats to practical cybersecurity frameworks tailored to the operational specifics of the industry. This literature review synthesizes recent research and industry reports to present a summary of the cybersecurity landscape in the O&G sector. It will identify common vulnerabilities, discuss the effectiveness of various cybersecurity measures, and propose strategic recommendations to enhance industry resilience against cyber threats.

II. BACKGROUND

Historically, the O&G sector has been slow to adopt technology [7], especially cybersecurity measures, compared to other industries [4]. This delay is partly due to its early reliance on isolated systems, which were presumed secure from cyber threats. However, as the industry adopted more integrated and interconnected IT and OT systems to enhance efficiency and production, the attack surface expanded, increasing the risk of cyber incidents [8]).

The evolution of cyber threats has been marked by increasing sophistication and frequency. Initially, most cyber threats were opportunistic, targeting low-hanging fruits like poorly secured networks and devices [9]. However, over time, attacks have become more sophisticated, involving specific threats such as advanced persistent threats (APTs), ransomware, and social engineering tactics. These tactics are specifically designed to penetrate the O&G sector's defenses, posing a significant risk not only to financial gain, but to critical national infrastructure (CNI), operations, health and the environment [1].

An Advanced Persistent Threat (APT) attack follows a systematic multi-stage process as shown in the figure 1. Attackers first conduct surveillance to gain access to a network, then establish and deepen their foothold. They manipulate system vulnerabilities to move laterally, searching for valuable data

and exploiting network weaknesses. Throughout, they deploy tools to maintain their presence discreetly, learning about the system's defenses and remaining undetected. The attack culminates with the attackers exerting control over systems to achieve their ultimate objectives, such as data theft or disruption of operations. This systematic approach underscores the need for comprehensive defense strategies [10].



Fig. 1: APT Execution process [10].

Regulative responses to these evolving threats have varied worldwide. US guidelines, such as those from the (NIST), help with risk assessment and management [11]. However, adherence to such policies is often voluntary, leading to differing levels of industry preparedness for security. Furthermore, the EU's Directive on Security of Network and Information Systems (NIS 2) seeks to raise cybersecurity standards in critical industries such as O&G [12], [13]. However, consistent implementation and enforcement are still needed.

The new trends show that more robust cybersecurity measures are becoming more accepted. More often, O&G companies contribute capital to cybersecurity technologies like intrusion detection strategies, network segmentation, and threat intelligence platforms [4]. Cybersecurity awareness training for employees to help them avoid human error and insider threats is also being promoted [6].

The interconnectedness of IT and OT networks creates a significant vulnerability, as cyberattacks on enterprise systems can lead to cascading disruptions in physical operations [14]. This phenomenon of 'indirect shutdowns' emphasizes the need for strategies like network uncoupling to enhance resilience against ransomware and hacktivism.

Cyber-Informed Engineering (CIE) grew out of a need to actively manage cybersecurity risks in critical infrastructure [15]. Integrated security management across the entire engineering

lifecycle is an essential need in the oil and gas sector, with interconnected OT systems vulnerable to cyber threats. With security in mind, CIE reduces vulnerabilities, attack surfaces, and impacts of cyber incidents by designing systems [16].

III. CYBERSECURITY CHALLENGES IN THE OIL AND GAS INDUSTRY

A. Technological Challenges

The O&G industry faces significant cybersecurity challenges because it adopts high-value infrastructure (see Figure 2). The main issues are the dependence on outdated systems, the inclusion of newer technologies like IoT and cloud services, as well as the associated risks.

Cybersecurity Challenges in the Oil and Gas Industry		
Technological Challenges	Organizational Challenges	Regulatory and Compliance Challenges
• Legacy Systems: Vulnerabilities in outdated technology. • Integration of New Technologies: Increased attack surface due to IoT and cloud services. • Associated Risks: Complexity of IT/OT convergence and heightened cyber threat landscape.	• Cybersecurity Culture: Leadership commitment needed for a security-centric culture. • Awareness and Training: Ongoing education essential for employee cybersecurity awareness. • Policy Implementation: Ensuring consistent enforcement across departments and regions.	• Overview of Regulations: Varied global, national, and regional standards. • Compliance Issues: Complexity, resource allocation, and adaptation to evolving laws. • Strategies for Managing Compliance: Dedicated teams, technology, and regular training recommended for effective compliance.

Fig. 2: Types of cybersecurity challenges in the O&G Industry

1) *Legacy Systems:* Legacy systems in the industry often comprise outdated technology that was not designed initially to confront current cybersecurity threats. These systems are prevalent in operational technology (OT) environments, including drilling, production, and transportation control systems [1]. The primary issues with legacy systems include outdated software that cannot be patched or updated, lack of compatibility with modern cybersecurity tools, and insufficient security controls [27]. This situation makes them particularly vulnerable to cyberattacks, which could lead to significant disruptions in operational processes and safety protocols.

2) *Integration of New Technologies:* The integration of IoT and cloud computing into the existing infrastructure of the industry brings about opportunities and risks. IoT devices, used for monitoring and controlling physical operations remotely, exponentially increase the number of network endpoints. These devices often lack robust built-in security, making them easy targets for cyber attackers [28]. Cloud services offer scalable data storage and processing capabilities, but they also introduce concerns such as data privacy, unauthorized access, and data integrity issues.

3) *Associated Risks:* The convergence of information and operational technologies creates a complex cybersecurity landscape. Cyber threats can now traverse from traditional IT

TABLE I: Selected Major Cybersecurity Attacks on the O&G Sector in the Last Decade

No.	Year	Target (Country)	Type of Attack	Outcome/Impact	Reference
1	2024	Ukraine's Largest Gas and Oil Company	Yet to be known	Website and call center operations were disrupted. The attack targeted IT systems, not the more sensitive industrial computers that control energy production and delivery	[17]
2	2023	Gas station chain (Canada)	Supply Chain Attack	Widespread outages and disruptions across Suncor's operations and systems.	[18]
3	2022	Oil Refining Ports and Storage Facilities (Europe)	Supply Chain Attack	This sophisticated attack shut down numerous ports, resulting in supply disruptions worldwide, with billions of Euros being lost in the process.	[19]
4	2021	Colonial Pipeline (USA)	Ransomware Attack	Led to a major pipeline shutdown, affecting fuel supply in the Eastern USA.	[20]
5	2020	Chevron Corporation	Virus Attack	Ekans ransomware attack targeted Chevron's Control Systems and Operational Technology	[21]
6	2019	Saudi Aramco	Computer Virus Attack	The cyber attacks on Saudi Aramco have disrupted oil production, exposed vulnerabilities in cybersecurity infrastructure	[22]
7	2018	Italian and Middle East Energy Industry	Malware Attack	Approximately 300 servers and approximately 100 personal computers, constituting a fraction of the aggregate count of approximately 4,000 machines, were rendered inoperable or significantly impaired.	[23]
8	2017	Petrochemical Plants (Saudi Arabia)	Triton/Trisis Malware	Intended to cause physical damage by targeting safety systems.	[24]
9	2016	Saudi Arabia	Shamoom Malware	The malware quickly wiped out data on all affected computers, leading to months-long recovery efforts.	[25]
10	2014	Companies in Norway's O&G Sector	Spear Phishing	Gained extensive network access, compromising sensitive operational data.	[26]
11	2012	Saudi Aramco (Saudi Arabia)	Shamoon Virus	Erased data on 35,000 computers, halting business operations.	[25]

environments, where data confidentiality and integrity are paramount, into OT environments, which prioritize system availability and safety [29]. This interconnectedness can lead to cascading effects from a single point of failure, significantly amplifying the potential impact of cyberattacks [30]. Additionally, the O&G industry is a prime target for nation-state actors and cybercriminals due to its economic importance, further elevating the risk profile [9].

B. Organizational Challenges

The O&G industry faces significant organizational cybersecurity challenges, which can undermine even the most robust technical defenses. A disconnect often exists between established policies and employee practices, highlighting the need for stronger leadership commitment to foster a culture where security is prioritized [31]. Additionally, training programs frequently fall short by being generic and not tailored to industry-specific needs, hindering their effectiveness in addressing evolving threats [1]. The consistent implementation and enforcement of policies across diverse operations, particularly in multinational corporations, is further complicated by varying regulatory environments and requirements [32]. These organizational shortcomings underscore the importance of addressing human factors and cultural aspects alongside technological advancements to ensure comprehensive cybersecurity in the O&G sector.

C. Regulatory and Compliance Challenges

The O&G industry is globally interconnected and strategically important, therefore subject to a web of national and regional cybersecurity regulations. Their compliance is challenging because organizations must navigate a maze of international, national, and industry standards. Cybersecurity regulations for the O&G industry aim to protect safety and environmental-critical operations from cyber attacks. Those regulations can be divided into mandatory government regulations and voluntary industry standards. [33].

- **International Standards:** The International Organization for Standardization (ISO) provides guidelines such as ISO/IEC 27001, which provides a structured system for information security management systems (ISMS) applicable across industries, including O&G [34]. The ISA/IEC 62443 standards offer a comprehensive framework for securing industrial automation and control systems. They cover various aspects of cybersecurity, enabling asset owners, system integrators, and product suppliers to enhance their systems' resilience against cyber threats [35].
- **National Regulations:** Different countries have their specific regulations and protocols. For example, the Transportation Security Administration (TSA) in the United States has guidelines specific to pipeline security, while the Federal Energy Regulatory Commission (FERC) oversees the electric sector. The NIS 2 Directive of the European Union seeks to enhance security measures across network and information systems [13], [36].

- **Regional Initiatives:** Regions may also have collective cybersecurity initiatives. For instance, the European Union Agency for Cybersecurity (ENISA) provides support for EU states in implementing the NIS Directive and enhancing their cybersecurity capabilities [37].

1) *Compliance Issues:* The complex and ever-changing nature of cybersecurity regulations presents significant challenges for organizations. The breadth and variability of these regulations can lead to ambiguity and inconsistency, especially for multinational corporations navigating diverse regulatory environments [38]. Compliance demands substantial resource allocation, including training for compliance officers, investment in technology, and regular audits, which can be particularly burdensome for smaller companies. Moreover, the dynamic nature of cybersecurity threats necessitates constant adaptation of policies and procedures, requiring companies to stay informed and agile to keep pace with new laws and amendments [39].

IV. CYBERSECURITY SOLUTIONS AND BEST PRACTICES

A. Technological Solutions

Security is a constantly evolving landscape. Figure 3 gives an overview of cybersecurity solutions and best practices for the oil & gas industry. Core technologies for securing digital assets include Intrusion Detector Systems (IDS), firewalls, and encryption methods. They constitute the core of security architectures and are crucial for mitigating risks of cyber threats [40]. There is a growing consensus within the industrial control systems (ICS) sector that vendors should prioritize the development of inherently more secure devices and protocols.

Cybersecurity Solutions and Best Practices		
Technological Solutions	Strategic Measures	Policy Adherence
1.IDS (NIDS, HIDS) 2.Firewalls 3.Encryption	1.Risk Frameworks (NIST, ISO 27001, CIS Controls) 2.Cyber Governance 3.Incident Response 4.Employee Training 5.Third-Party Risk Management 6.Continuous Improvement	1.Data Regulations (GDPR, CCPA, HIPAA) 2.Industry Standards (PCI DSS, HIPAA) 3.Security Policies 4.Compliance Monitoring 5.Vendor Management 6.Training & Awareness

Fig. 3: Cybersecurity Solutions and Best Practices in the O&G Industry

1) *Intrusion Detection Systems (IDS):* IDS can detect unwanted access or unusual network traffic behaviors. There are two types: Network-based Intrusion Detection Systems (NIDS) & Host Based Intrusion Detection Systems (HIDS). NIDS

monitors network traffic for anomalies; HIDS monitors system calls and files system changes on a machine-by-machine basis. [41]. Real-world studies have demonstrated the effectiveness of IDS.

2) *Firewalls:* Firewalls keep reliable and untrusted networks apart and control network traffic in both directions according to established security protocols. They block unwanted access to network resources. Firewalls may be hardware or software-based, and each kind offers various levels of control over security [42]. Their research paper [43] discusses the challenges caused by modern network security threats and advocates for the adoption of Next Generation Firewalls (NGFWs) to counter cybersecurity risks effectively. While firewalls focus on monitoring and protecting data, unidirectional technology provides a fundamental level of protection by preventing any data return from less secure to more secure environments. They are the most widely deployed solution to segment IT and OT networks.

3) *Encryption:* Encryption technology is essential for protecting the confidentiality of data. It converts easily readable data into a coded format that can only be accessed or manipulated once it has been deciphered. Encryption is used in various forms, such as symmetric-key encryption, asymmetric-key encryption, and hashing, based on the specific criteria of the application [44]. A notable application of encryption is in securing communications over the Internet, as discussed by [45], who explains the operations of cryptographic techniques like AES and RSA in ensuring data security.

4) *Hardware-Enforced Remote Access for O&G:* Hardware-enforced network segmentation enhances security for remote access in OT environments like O&G, using unidirectional gateways or data diodes to create physical barriers against unauthorized access. This allows secure remote monitoring and support without exposing critical systems to the internet, addressing the industry’s unique security challenges and enabling remote connectivity benefits.

B. Strategic and Organizational Measures

In today’s connected digital landscape, organizations must adopt strategic and organizational measures to mitigate cybersecurity risks effectively. These measures encompass a range of initiatives aimed at enhancing the resilience of systems, protecting sensitive data, and fortifying defenses against evolving threats.

1) *Risk Management Frameworks:* Adopting resilient risk management frameworks like the NIST Cybersecurity Framework, ISO 27001, or CIS Controls enables organizations to systematically identify, assess, and mitigate cybersecurity risks [46]. By adhering to these principles, firms can implement a systematic approach to cybersecurity governance and guarantee comprehensive risk mitigation techniques.

2) *Cybersecurity Governance:* Clear cybersecurity governance structures are crucial for effective risk oversight and decision-making. Define roles, responsibilities, & reporting

mechanisms, & integrate cybersecurity into policies and procedures. A strong governance drives accountability & creates a culture of cybersecurity awareness in the organization.. [47].

CIE enhances oil and gas cybersecurity by integrating security from the start, reducing vulnerabilities and attack surfaces, and improving resilience and situational awareness. This leads to cost-effective security, regulatory compliance, and a skilled workforce, safeguarding critical infrastructure and sensitive data in the face of evolving threats [15].

3) *Incident Response Planning*: Develop and test incident response procedures frequently to minimize the impact of cybersecurity incidents. Organizations need predefined steps for detecting, responding to, and recovering from breaches. Proactive preparation reduces financial, reputational, and operational effects of cyber incidents [48].

4) *Employee Training and Awareness*: Human mistakes continue to be a substantial factor in cybersecurity issues. Offering extensive training and awareness initiatives enables staff to acquire the necessary information and abilities to identify and address security issues efficiently. Training sessions, simulated phishing exercises, and awareness campaigns can empower staff to protect company security actively [49].

5) *Third-Party Risk Management*: As enterprises rely more on third-party vendors, managing associated cybersecurity risks is crucial. Effective risk mitigation involves due diligence, regular security assessments, and contractual obligations for security standards to protect organizational assets. [50].

6) *Continuous Monitoring and Improvement*: Cybersecurity is a perpetual undertaking that necessitates constant surveillance, assessment, and enhancement. By utilizing advanced technologies like Security Information and Event Management (SIEM) systems, threat intelligence platforms, and vulnerability management tools, enterprises may promptly identify and address emerging threats as they occur [51].

C. Policy and Regulatory Adherence

Policies and regulations set standards, procedures, and controls for protecting information assets and ensuring system and network integrity [52].

1) *Vendor Compliance Management*: Organizations engaging third-party vendors/service providers must ensure their cybersecurity practices meet regulatory and industry standards. Due diligence assessments, including security evaluations/contractual obligations, reduce the risk of violating regulations and security breaches due to vendor relationships [53].

2) *Industry-Specific Standards*: Different industries have tailored cybersecurity standards and regulations to their risk profiles and operational requirements. For example, Financial firms must abide by payment card Industry Data Security Standard PCI DSS to protect Payment Card transactions. With industry standards, organizations can meet sector-specific threats and regulatory expectations [54].

3) *Security Policy Development*: Thorough cybersecurity policies and procedures are the first step to developing a solid

security posture. Such policies should cover data classification/access control/incident response / permissible usage of IT resources / etc. By clearly defining security expectations and responsibilities, organizations can establish a baseline for security practices and enable consistent enforcement across the enterprise. [55].

4) *Regulatory Reporting and Compliance Monitoring*: Organizations need mechanisms for monitoring regulatory changes, assessing compliance requirements, and reporting regulatory adherence [56] might mean regular audits, risk evaluations & compliance inspections for deficiencies & conformity with evolving regulatory requirements.

5) *Training and Awareness*: Employee training and awareness programs play a crucial role in promoting regulatory compliance and fostering a culture of cyber security within the organization. Training workshops on relevant regulations, policies, and best practices empower employees to understand their compliance obligations and contribute to maintaining a secure environment for sensitive information [57].

6) *Data Protection Regulations*: Respecting Data Protection rules such as the General Data Protection Regulation and California Consumer Privacy Act is fundamental for organizations handling personal Data [58]. To meet O&G rules, appropriate technical and organizational procedures are needed to protect data confidentiality, integrity, and availability, as well as transparency and accountability for data processing activities.

V. FUTURE TRENDS

A rapidly evolving cybersecurity domain requires enterprises to predict trends and take proactive steps to deal with emerging threats. Various significant trends and orientations are anticipated to influence the future of cybersecurity, as shown in figure 4. In the realm of cybersecurity, proactive solutions are paramount. Government regulations and standards alone cannot safeguard critical infrastructure. Instead, a multi-layered defense approach is essential.

Proactive cybersecurity necessitates a "Secure by Design" philosophy, where security is ingrained into systems from their inception. This includes utilizing improved ICS devices, robust protocols, and unidirectional communication solutions to prevent unauthorized access and mitigate risks. Moreover, well-engineered networks with strong segmentation serve to limit the impact of breaches and contain potential threats. Crucially, integrating cybersecurity considerations into the engineering design phase (CIE) ensures that security measures are inherently part of the system, not an afterthought.

AI and Machine Learning are revolutionizing cybersecurity by enabling rapid analysis of massive datasets, identifying trends, and detecting threats in real-time, helping organizations stay ahead of sophisticated attackers [59]. Simultaneously, the traditional perimeter-based security model is giving way to Zero Trust Architecture, which emphasizes continuous identity verification, privilege controls, and micro-segmentation to mitigate insider threats and unauthorized access [60]. Furthermore,

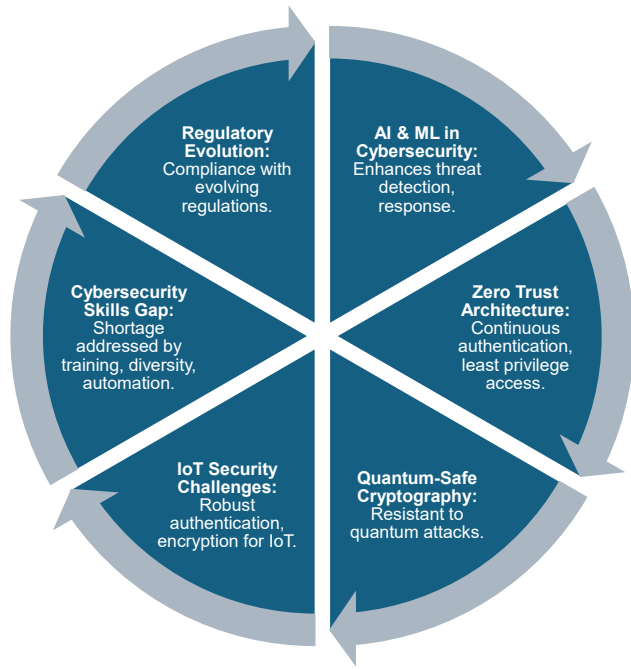


Fig. 4: Future Trends and Directions For Cybersecurity in the O&G Industry

the rise of quantum computing necessitates the development of quantum-safe cryptography to protect data from future threats, while the proliferation of IoT devices demands robust authentication, encryption, and device management to secure their expanding ecosystems [61], [62].

The proliferation of IoT devices introduces significant security challenges, necessitating robust authentication, encryption, and device management to protect these vulnerable ecosystems. Simultaneously, the persistent shortage of skilled cybersecurity professionals poses a major obstacle, requiring collaborative efforts from academia, industry, and government to bridge the gap [63]. Furthermore, the evolving landscape of cybersecurity regulations demands organizations remain vigilant and proactive in ensuring compliance to avoid potential penalties.

VI. CONCLUSION

O&G has significant cybersecurity challenges because of their importance to the global economy and evolving cyber threats. Key vulnerability areas in the industry are reviewed, along with mitigation options and strategic recommendations. Technologies, regulatory and organizational challenges abound. Legacy systems, modern digital technologies, and various standards create a complicated cybersecurity landscape. The industry responded with technological solutions like intrusion detection systems and strategic measures like risk management frameworks. Yet improvement is necessary. New trends such as AI, Zero Trust Architecture & quantum-safe cryptography open up new defenses. The cybersecurity skills gap, IoT ecosystem security, and adaptation to changing regulatory

environments are key focus areas. Such a proactive and holistic approach involving collaboration between stakeholders is necessary to protect the industry assets and ensure their security and reliability. Cyber risk mitigation and opportunities for improvement require ongoing research and policy development.

DECLARATION

In preparing this work, the authors utilized ChatGPT4.0 and Grammarly to enhance the language quality. Following the use of these tools, the authors carefully reviewed and revised the content, assuming complete responsibility for the final publication.

REFERENCES

- [1] I. Progoulakis, N. Nikitakos, P. Rohmeyer, B. A. Bunin, D. Dalaklis, and S. Karamperidis, "Perspectives on cyber security for offshore oil and gas assets," *Journal of Marine Science and Engineering*, 2021.
- [2] B. Dickinson and D. Wilkinson, "Securing industrial systems in a digital world," *The APPEA Journal*, 2019.
- [3] S. Badi and H. Said, "Digital transformation and the cybersecurity of infrastructure systems in the oil and gas sector," in *Routledge Handbook of Planning and Management of Global Strategic Infrastructure Projects*, pp. 401–429, Routledge, 2020.
- [4] H. Imran, M. Salama, and C. Turner, "Digitization, cybersecurity and risk management in the oil and gas sector in the post covid world: A systematic review," *Human-Centered Design and User Experience*, 2023.
- [5] F. R. Bechara and S. B. Schuch, "Cybersecurity and global regulatory challenges," *Journal of Financial Crime*, vol. 28, no. 2, pp. 359–374, 2021.
- [6] M. Carmony, *Analyzing Cybersecurity Risks Within Critical Infrastructures: Threats and Challenges in the Oil and Gas Industry*. PhD thesis, Utica College, 2020.
- [7] T. Nguyen, R. Gosine, and P. J. Warrian, "A systematic review of big data analytics for oil and gas industry 4.0," *IEEE Access*, vol. 8, pp. 61183–61201, 2020.
- [8] *Security Implications of IIoT Architectures for Oil & Gas Operations*, vol. Day 1 Tue, April 20, 2021 of *SPE Western Regional Meeting*, 04 2021.
- [9] G. Stergiopoulos, D. Gritzalis, and E. Limnaios, "Cyber-attacks on the oil & gas sector: A survey on incident assessment and attack patterns," *IEEE Access*, vol. 8, pp. 128440–128475, 2020.
- [10] P. Chen, L. Desmet, and C. Huygens, "A study on advanced persistent threats," in *Communications and Multimedia Security: 15th IFIP TC 6/TC 11 International Conference, CMS 2014, Aveiro, Portugal, September 25–26, 2014. Proceedings 15*, pp. 63–72, Springer, 2014.
- [11] H. Hellmann, "Acknowledging the threat: Securing united states pipeline scada systems," *The Energy Law Journal*, vol. 36, p. 157, 2015.
- [12] D. Markopoulou, V. Papakonstantinou, and P. De Hert, "The new EU cybersecurity framework: The NIS Directive, ENISA's role and the General Data Protection Regulation," *Computer Law & Security Review*, vol. 35, no. 6, p. 105336, 2019.
- [13] P. Eckhardt and A. Kotovskaia, "The eu's cybersecurity framework: the interplay between the cyber resilience act and the nis 2 directive," *International Cybersecurity Law Review*, vol. 4, no. 2, pp. 147–164, 2023.
- [14] A. F. Ginter, *Cyber Perimeters for Critical Infrastructures*, pp. 67–100. Cham: Springer International Publishing, 2017.
- [15] R. S. Anderson, J. Benjamin, V. L. Wright, L. Quinones, and J. Paz, "Cyber-informed engineering," tech. rep., Idaho National Laboratory (INL), Idaho Falls, ID (United States), 2017.
- [16] API, "Api | cybersecurity." <https://www.api.org/news-policy-and-issues/cybersecurity>. (Accessed on 09/16/2024).
- [17] "Ukraine's largest gas and oil company under cyberattack." <https://www.kyivpost.com/post/27165>. (Accessed on 05/02/2024).
- [18] "Canadian oil giant suncor confirms cyberattack after countrywide outages." <https://therecord.media/canadian-oil-giant-suncor-cyberattack>. (Accessed on 05/02/2024).

- [19] "Cyber threats for the oil and gas industry | meriplex." <https://meriplex.com/cyber-threats-for-the-oil-and-gas-industry/#:~:text=2022%20European%20Oil%20Refining%20Ports,Refining%20Ports%20and%20Storage%20Facilities.> (Accessed on 05/02/2024).
- [20] J. Beerman, D. Berent, Z. Falter, and S. Bhunia, "A review of colonial pipeline ransomware attack," in *2023 IEEE/ACM 23rd International Symposium on Cluster, Cloud and Internet Computing Workshops (CC-GridW)*, pp. 8–15, IEEE, 2023.
- [21] "Igniting attacks: Cybersecurity in the oil and gas industry." <https://www.sangfor.com/blog/cybersecurity/cybersecurity-in-the-oil-and-gas-industry#:~:text=The%20Chevron%20Corporation%2C%20one%20of,vulnerability%20in%20its%20VPN%20software.> (Accessed on 05/02/2024).
- [22] "Saudi aramco attacks could predict cyber warfare from iran." <https://www.cnn.com/2019/09/21/saudi-aramco-attacks-could-predict-cyber-warfare-from-iran.html>. (Accessed on 05/02/2024).
- [23] "New shamoon malware variant targets italian oil and gas company." <https://thehacknews.com/2018/12/shamoon-malware-attack.html>. (Accessed on 05/02/2024).
- [24] "Triton is the world's most murderous malware, and it's spreading | mit technology review." <https://www.technologyreview.com/2019/03/05/103328/cybersecurity-critical-infrastructure-triton-malware/>. (Accessed on 05/02/2024).
- [25] "5 big cyberattacks in oil and gas." <https://www.oilandgasiq.com/digital-transformation/articles/5-big-cyber-security-attacks-in-oil-and-gas.> (Accessed on 05/02/2024).
- [26] "Oil and energy industry in norway is under attack - cyber defense magazine." <https://www.cyberdefensemagazine.com/oil-and-energy-industry-in-norway-is-under-attack/>. (Accessed on 05/02/2024).
- [27] F. Khorrami, P. Krishnamurthy, and R. Karri, "Cybersecurity for control systems: A process-aware perspective," *IEEE Design & Test*, vol. 33, pp. 75–83, 2016.
- [28] J. Wurm, K. Hoang, O. Arias, A. Sadeghi, and Y. Jin, "Security analysis on consumer and industrial iot devices," *2016 21st Asia and South Pacific Design Automation Conference (ASP-DAC)*, pp. 519–524, 2016.
- [29] S. E. McLaughlin, C. Konstantinou, X. Wang, L. Davi, A. Sadeghi, M. Maniatakos, and R. Karri, "The cybersecurity landscape in industrial control systems," *Proceedings of the IEEE*, vol. 104, pp. 1039–1057, 2016.
- [30] R. Kour, "Cybersecurity issues and challenges in industry 4.0," *Research Anthology on Cross-Industry Challenges of Industry 4.0*, 2020.
- [31] R. Rob, T. Tural, G. McLorn, A. O. Sheikh, and A. Hassan, "Addressing cyber security for the oil, gas and energy sector," *2014 North American Power Symposium (NAPS)*, pp. 1–8, 2016.
- [32] H. Bruijn and M. Janssen, "Building cybersecurity awareness: The need for evidence-based framing strategies," *Gov. Inf. Q.*, vol. 34, pp. 1–7, 2017.
- [33] T. R. Wanasinghe, L. Wroblewski, B. K. Petersen, R. Gosine, L. James, O. de Silva, G. Mann, and P. J. Warrian, "Digital twin for the oil and gas industry: Overview, research trends, opportunities, and challenges," *IEEE Access*, vol. 8, pp. 104175–104197, 2020.
- [34] G. Disterer, "Iso/iec 27000, 27001 and 27002 for information security management," *Journal of Information Security*, vol. 2013, pp. 92–100, 2013.
- [35] International Society of Automation, "Isa/iec 62443 series of standards." <https://www.isa.org/standards-and-publications/isa-standards/isa-iec-62443-series-of-standards>, n.d. Retrieved September 16, 2024.
- [36] R. J. Anderson and S. Fuloria, "Security economics and critical national infrastructure," pp. 55–66, 2009.
- [37] D. Markopoulou, V. Papakonstantinou, and P. Hert, "The new eu cybersecurity framework: The nis directive, enisa's role and the general data protection regulation," *Comput. Law Secur. Rev.*, vol. 35, p. 105336, 2019.
- [38] R. Quandt, "The welfare effects of complexity and variability in regulations," *Economics Letters*, vol. 13, pp. 259–261, 1983.
- [39] G. Culot, F. Fattori, M. Podrecca, and M. Sartor, "Addressing industry 4.0 cybersecurity challenges," *IEEE Engineering Management Review*, vol. 47, pp. 79–86, 2019.
- [40] V. Anand, "Intrusion detection: Tools, techniques and strategies," *Proceedings of the 42nd annual ACM SIGUCCS conference on User services*, 2014.
- [41] L. Dali, K. Abouelmehdi, A. Bentajer, H. Elsayed, E. Abdelmajid, and B. Abderahim, "A survey of intrusion detection system," *2015 2nd World Symposium on Web Applications and Networking (WSWAN)*, pp. 1–6, 2015.
- [42] S. Bellovin and B. Cheswick, "Network firewalls," *IEEE Communications Magazine*, vol. 32, pp. 50–57, 1994.
- [43] R. S. Yadav and P. Likhari, "Firewall: A vital constituent of network security," in *Information Technology Security: Modern Trends and Challenges*, pp. 47–67, Springer, 2024.
- [44] B. Kaliski, "A survey of encryption standards," *IEEE Micro*, vol. 13, pp. 74–81, 1993.
- [45] S. Fatima, T. Rehman, M. Fatima, S. Khan, and M. A. Ali, "Comparative analysis of aes and rsa algorithms for data security in cloud computing," *Engineering Proceedings*, vol. 20, no. 1, p. 14, 2022.
- [46] U. Saritac, X. Liu, and R. Wang, "Assessment of cybersecurity framework in critical infrastructures," *2022 IEEE Delhi Section Conference (DELCON)*, pp. 1–4, 2022.
- [47] J. Loonam, J. B. Zwieglelaar, P. V. Kumar, and C. Booth, "Cyber-resiliency for digital enterprises: A strategic leadership perspective," *IEEE Transactions on Engineering Management*, vol. PP, pp. 1–14, 2020.
- [48] A. Ahmad, J. Hadgkiss, and A. B. Ruighaver, "Incident response teams - challenges in supporting the organisational security function," *Comput. Secur.*, vol. 31, pp. 643–652, 2012.
- [49] M. Sas, G. Reniers, K. Ponnet, and W. Hardyns, "The impact of training sessions on physical security awareness: Measuring employees' knowledge, attitude and self-reported behaviour," *Safety Science*, vol. 144, p. 105447, 2021.
- [50] A. Ganin, P. Quach, M. Panwar, Z. A. Collier, J. Keisler, D. Marchese, and I. Linkov, "Multicriteria decision framework for cybersecurity risk assessment and management," *Risk Analysis*, vol. 40, 2020.
- [51] G. G. Granadillo, S. Zarzosa, and R. Díaz, "Security information and event management (siem): Analysis, trends, and usage in critical infrastructures," *Sensors (Basel, Switzerland)*, vol. 21, 2021.
- [52] M. Cavallari, "A conceptual analysis about the organizational impact of compliance on information security policy," pp. 101–114, 2012.
- [53] O. F. Keskin, K. M. Caramancion, I. Tatar, O. Raza, and U. Tatar, "Cyber third-party risk management: A comparison of non-intrusive risk scoring reports," *Electronics*, 2021.
- [54] S. Yulianto, C. Lim, and B. Soewito, "Information security maturity model: A best practice driven approach to pci dss compliance," *2016 IEEE Region 10 Symposium (TENSYP)*, pp. 65–70, 2016.
- [55] R. J. Harknett and J. A. Stever, "The new policy world of cybersecurity," *Public Administration Review*, vol. 71, pp. 455–460, 2011.
- [56] M. Siponen, M. Mahmood, and S. Pahlila, "Employees' adherence to information security policies: An exploratory field study," *Inf. Manag.*, vol. 51, pp. 217–224, 2014.
- [57] T. R. Peltier, "Implementing an information security awareness program," *Information Systems Security*, vol. 14, pp. 37 – 49, 2005.
- [58] J. Krystlik, "With gdpr, preparation is everything," *Computer Fraud & Security*, vol. 2017, pp. 5–8, 2017.
- [59] J. P. Bharadiya, "Ai-driven security: How machine learning will shape the future of cybersecurity and web 3.0," *American Journal of Neural Networks and Applications*, vol. 9, no. 1, pp. 1–7, 2023.
- [60] V. Stafford, "Zero trust architecture," *NIST special publication*, vol. 800, p. 207, 2020.
- [61] S. Y. Yan and S. Y. Yan, "Quantum safe cryptography," *Cybercryptology: Applicable Cryptography for Cyberspace Security*, pp. 399–412, 2019.
- [62] U. Khalil, . Mueen-Uddin, O. Malik, and S. Hussain, "A blockchain footprint for authentication of iot-enabled smart devices in smart cities: State-of-the-art advancements, challenges and future research directions," *IEEE Access*, vol. 10, pp. 76805–76823, 2022.
- [63] S. Furnell, "The cybersecurity workforce and skills," *Comput. Secur.*, vol. 100, p. 102080, 2021.